2017-2018 – DEEQA – TSE – Advanced Macroeconomics

Assignment 2

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Assume we have a system of n linear difference equations

$$\mathbb{E}_t Y_{t+1} = \mathrm{A}\, Y_t + \mathrm{B}\, V_t$$

$$V_t = \rho V_{t-1} + \epsilon_t$$

where $\mathbb{E}_t Y_{t+1}$ represents the expectation of Y_{t+1} given the information set available at time t, V_t is a $(k \times 1)$ vector of exogeneous driving processes and Y_t is a $(n \times 1)$ vector where the first ℓ variables are state variables and the remaining $n - \ell$ are jump/free variables.

1. Write a computer program (e.g. MATLAB) that takes as input the above system of expectational difference equations (ie, the matrix A, B and ρ above)
2. The program should then provide the state space representation of the solution, and give impulse response of different variables to shocks. It should at some point check that the equilibrium is determinate.
3. Also get the program to generate artificial data with the model (starting from the zero steady state), and calculate variance-covariance matrices of the variables
4. Try also to derive theoretical moments (as functions of the parameters of the state space representation)